

Immunopathology of Urticaria

Tumor necrosis factor- α (TNF- α) has been detected on epidermal keratinocytes in both lesional and non-lesional skin biopsy specimens from patients with chronic idiopathic urticaria.

In chronic idiopathic urticaria, CD11b and CD18-positive cells are found surrounding blood vessels in the superficial and deep dermis. Direct immunofluorescence testing for immunoglobulins and complement components (such as C3) is typically negative.

Major basic protein (MBP) and eosinophil cationic protein (ECP), both derived from eosinophil granules, are present perivascularly and diffusely distributed in the dermis in various forms of urticaria, including:

- Acute urticaria
- Chronic idiopathic urticaria
- Delayed pressure urticaria
- Cholinergic urticaria
- Solar urticaria

In chronic idiopathic urticaria, free eosinophil granules are more abundant in wheals lasting longer than 24 hours compared to those of shorter duration. The secreted forms of ECP and eosinophil-derived neurotoxin (EDN) are found in higher levels in skin biopsy specimens from patients with chronic urticaria who lack autoantibodies, compared to those who have them.

Adhesion molecules including P-selectin, E-selectin, intercellular adhesion molecule 1 (ICAM-1), and vascular cell adhesion molecule 1 (VCAM-1) are upregulated on vascular endothelium in patients with chronic idiopathic urticaria and dermographism. Additionally, major histocompatibility complex (MHC) class II antigens are overexpressed on endothelial cells. Peripheral blood lymphocytes in these patients show increased expression of CD40 ligand and Bcl-2, suggesting enhanced autoimmune activity.

In papular urticaria, histopathologic findings include epidermal thickening with intercellular edema and lymphocytic infiltration. The dermis shows edema accompanied by an inflammatory infiltrate composed of T lymphocytes,

macrophages, eosinophils, and neutrophils—B lymphocytes and deposition of immunoglobulins, fibrin, or C3 are absent.

Treatment

Acute Urticaria

First-line therapy for acute urticaria consists of nonsedating antihistamines, as outlined in clinical guidelines. However, when symptoms are severe or generalized, or when angioedema is present, additional measures may be required.

If nonsedating antihistamines are insufficient, sedating antihistamines such as **hydroxyzine** or **diphenhydramine** (25–50 mg four times daily) can be used. A short course of systemic corticosteroids may also be beneficial—for example, **60–80 mg/day of prednisone for 3 days**, followed by a gradual taper of **5–10 mg/day**.

Epinephrine is indicated for severe manifestations such as generalized urticaria, intense pruritus, rapidly progressing angioedema, or laryngeal edema. It provides rapid relief of life-threatening symptoms associated with anaphylaxis. Note that edema of the posterior tongue or pharynx may mimic laryngeal edema but does not always pose the same risk.

General Principles

The optimal management of urticaria and angioedema involves identifying and eliminating the underlying cause. However, many cases of acute urticaria resolve spontaneously, and individuals often self-manage without medical consultation.

For chronic urticaria, treatment focuses on symptomatic control. Physicians should offer not only pharmacologic therapy but also emotional support and reassurance. Patients with chronic idiopathic urticaria frequently report that **pruritus** and the **unpredictable nature of flare-ups** are the most distressing aspects. Facial angioedema is particularly troubling, and swelling of

the tongue or pharynx may be perceived as life-threatening—though this differs from true laryngeal edema seen in anaphylaxis, C1 esterase inhibitor (C1 INH) deficiency, or ACE inhibitor reactions.

Chronic urticaria significantly impacts quality of life, with patients reporting: -
Sleep disturbances

- Fatigue
- Social isolation
- Emotional distress
- Impairment in work, home, social, and sexual functioning

Studies have shown a correlation between disease severity and **depression**. Among subtypes, **delayed pressure urticaria** and **cholinergic urticaria** are associated with the greatest reduction in quality of life. Cholinergic urticaria can interfere with exercise and sexual activity.

Although urticaria can be frustrating for both patients and clinicians, most individuals achieve acceptable symptom control even without identifying a specific cause. In some patients, avoidance of **aspirin** and other **NSAIDs** is important, as these agents can exacerbate symptoms.